

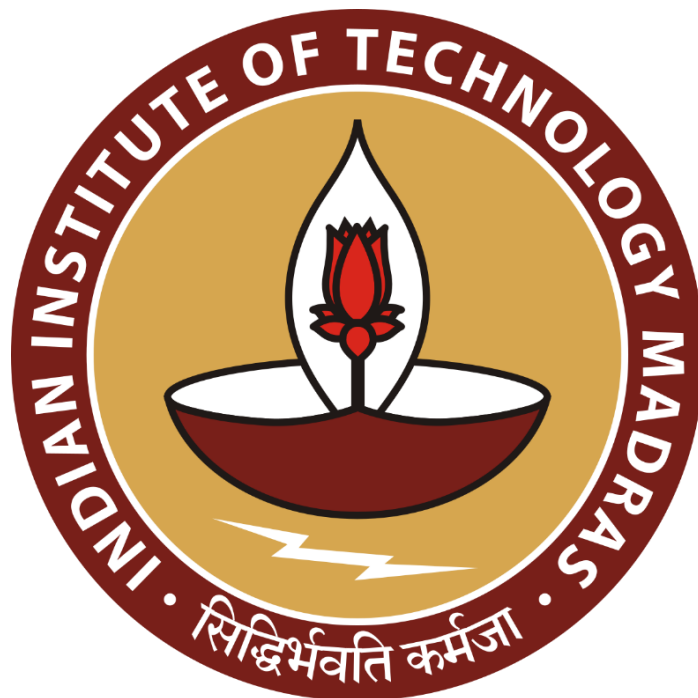
Quiz Master - V2

MAD 2 Project Report

Submitted by

Name: **INDAL BIND**

Roll number: **23f1002973@ds.study.iitm.ac.in**



IITM Online BS Degree Program,

Indian Institute of Technology, Madras, Chennai

Tamil Nadu, India, 600036

Table of Contents

- Description
- Technologies Used
- DB schema design
- API Design
- Architecture and Feature
- Video

Description

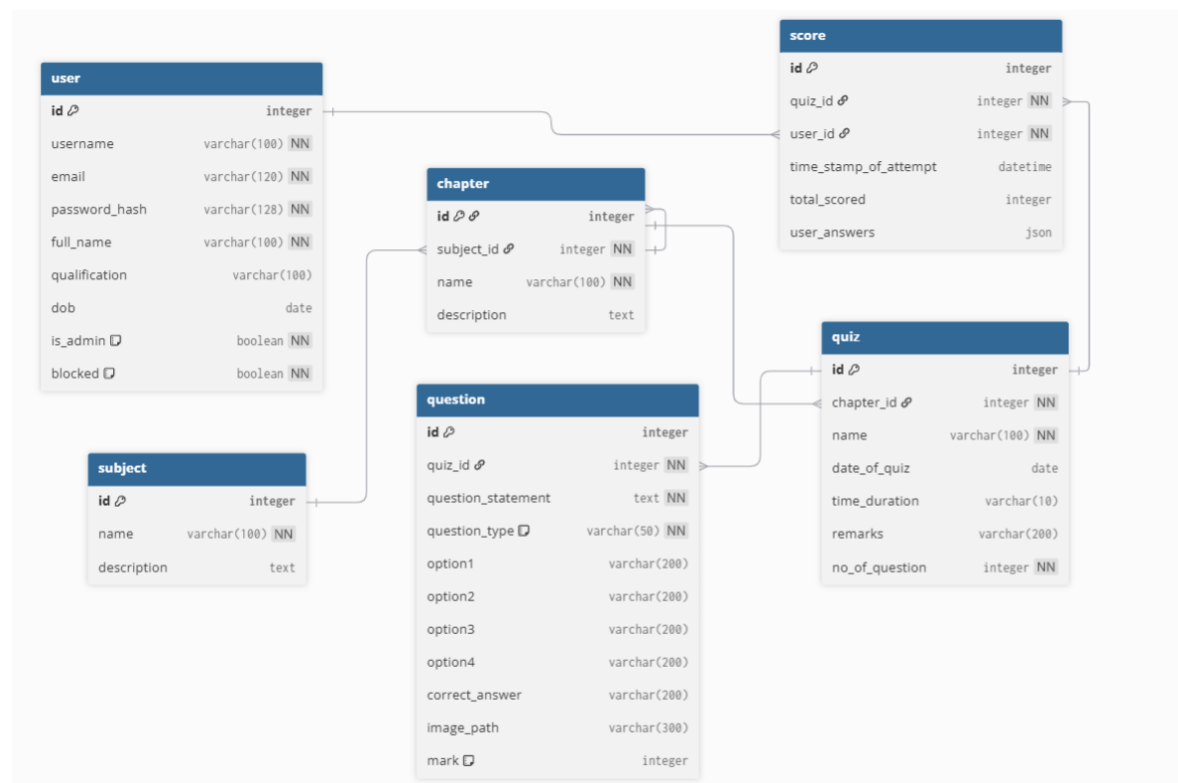
QuizMaster is a multi-user web application designed to help students effectively prepare for exams. In this app only one admin and many user, Users can join the platform, attempt quizzes chapter-wise or subject-wise, and receive detailed performance reports after each attempt.

Technologies Used

- **Backend:** Flask, SQLAlchemy, Redis, Celery, flask-JWT-Extended
- **Frontend:** Javascript , HTML, CSS, VueJS 2, Bootstrap, font-awesome, Axios, Chart.js
- **Database:** SQLite
- **Libraries:** JWT for security, Flasgger for API documentation, ChartJS for data visualization

DB Schema Design

Data base are in the link of :- <https://dbdiagram.io/d/6868252ef413ba35085b9732>



API Design

I make many api for communicate between the frontend and backend **api are Covered**

User: registration, login, dashboard

Subject: create, view, update, delete (admin only)

Chapter: create, view by subject, update, delete (admin only)

Quiz: create, list quizzes, view quiz details, delete (admin only)

Question: add/edit/remove questions within quizzes (admin only)

Score: submit quiz responses, fetch quiz result/report (user)

Reminder and mail : quiz reminders or new quiz announcements (via Redis + Celery)

The Api endpoint include

- Request method(get,post,put,delete)
- Input schema (like quiz creation details)
- Authentication
- Joining the table
- Response formate with success or error
- Giving the response in json

Architecture and Features

The project is organized as there is **Backend** folder which store the backend thing and **frontend** folder which include the frontend and inside the frontend the component(page) are in the static folder and the html file are in the frontend folder and every thing in the static like component, css, image and router , main app.js , so the controller are in the frontend folder of static folder. And the main app.py which run that are outside.

The mostly feature in the app where user can able to attempt the quiz based on subject and default feature is like user can login, register and the core feature are like

- Admin Panel (where admin can create quiz, delete, etc)
- User Panel
- Quiz Attempt
- Mail send
- Daily report
- Blocking user, deleting the user

Video

Link for video :-

<https://drive.google.com/file/d/1bHyO7TSG8TguOz4UPm9BS28Ny3GQTCsV/view?usp=sharing>

Thanks...